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# Drag force on a microrobot propelled through blood

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Controlling microrobot locomotion in vessels and capillaries is crucial for precise drug delivery and minimally invasive surgeries. However, this is challenging due to the complex interactions with red blood cells (RBCs) and the difficulty navigating within the dense environment. Here, we construct a numerical framework to evaluate the relative resistance coefficient ($C_r^*$) of a microrobot propelled through RBC suspensions. Our experiments validate the numerical results. We find that $C_r^*$ increases for smaller microrobots and higher hematocrit levels, while magnetic force strength weakly impacts $C_r^*$. $C_r^*$ is smaller than the resistance coefficient of a macroscale robot estimated from the apparent viscosity of the RBC suspension. The aspect ratio of a prolate ellipsoidal microrobot influences $C_r^*$ along its long-axis direction. Additionally, machine learning accurately predicts $C_r^*$. These insights could enhance the design and control of microrobots for medical applications.

Amicrorobot is a micro-scale robot that performs a specific medical task[1–5], such as drug delivery[6–8], in vivo imaging[9–11], or minimally invasive surgery[12–14]. During these applications, microrobots (hereinafter referred to as robots) must reach a certain region of the body, via blood vessels and capillaries[15–18]. Robot locomotion is affected by various complex physicochemical factors that may cause robots to unexpectedly change course or aggregate[19–21]. Despite notable recent progress, the propulsion capabilities of the majority of microrobots have remained confined to homogeneous Newtonian fluids and ideal environments[22–24]. However, blood, a medically important biological fluid, presents a complex and heterogeneous medium, characterized by intricate rheological properties profoundly influenced by the tangible blood components[23,25]. These inherent complexities pose substantial challenges, constraining the efficacy and control accuracy of microrobots propelled in blood[24,26]. RBCs play an extremely significant role in the blood, constituting about 99% of tangible blood components; the RBC volume fraction is 40-45%[27–31]. Indeed, blood, as the hemodynamic environment for robot migration, can be viewed as a suspension of red blood cells (RBCs)[27]. Consequently, microrobots encounter limitations rooted in precision, size constraints, and intricate hemodynamic complexities, all stemming from the influence of RBCs. Hence, there is an urgent imperative to elucidate microrobot dynamics in RBC suspension to optimize microbot locomotion parameters and to develop microrobots capable of autonomous navigation and precise control while delivering dependable long-term retention capabilities.

The key equation of robot migration is the Stokes' drag $\mathbf{F} = -6\pi\mu a_s\mathbf{v}$, where $\mathbf{F}$ is the drag force, $\mu$ is the viscosity, $a_s$ is the radius of the spherical robot, and $\mathbf{v}$ is the robot velocity. The Stokes' drag equation serves as a universal tool for predicting essential parameters like swimming speed, drag force, and drag torque for microrobots. This equation holds for Stokes flow (such as during robot migration), where the effect of inertia is negligible compared to the viscous effect. When a robot is sufficiently large compared to suspended particles, the suspension can be regarded as a homogeneous fluid and the drag force can be evaluated using the apparent viscosity $\eta$. In 1956, Einstein[32] analytically derived the apparent viscosity of a dilute suspension of rigid spheres as $\eta_d = \mu(1 + 2.5\phi_r)$, where $\phi_r$ is the volume fraction of rigid spheres. Batchelor and Green[33] extended this research to semi-dilute suspensions. The apparent viscosities of concentrated suspensions of rigid spheres have been intensively investigated in terms of yield stress, shear-thinning, shear-thickening, thixotropy, and rheopex[34–36]. Computer simulations have greatly improved our understanding of apparent viscosity. Stokesian dynamics[37], dissipative particle dynamics[38], lattice Boltzmann method[39], and Lagrange multiplier fictitious domain method[40] have all been used to explore the apparent viscosities of dilute-to-dense regimes.

When the robot is comparable in size to suspended particles, Stokes' drag calculated using the apparent viscosity is no longer applicable. The drag force is now a function of the hematocrit, the size of the robot relative to that of the suspended particles, and the ratio between the thrust of the robot and elastic force acting on the RBC membrane[41–44]. Thus, the resistance force can be obtained only via careful calculation of the many-body interactions between particles at the microscale. RBC mechanics have been investigated by many researchers over the past 50 years from theoretical, experimental, and computational perspectives[28,30,31]. As a constitutive equation for RBC membranes, the Skalak law[45], which considers shear elasticity and area incompressibility, has been widely used. In terms of simulations, the boundary element method (BEM) is one of the most accurate for simulating RBCs in Stokes flow because it takes account of the discontinuity of the

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hydrodynamic stress tensor across the membrane[46]. Although many studies provide fundamental knowledge on the drag force in blood, the role played by RBCs in robot migration has received little attention[47–52].

Here, we investigate the locomotion of a single robot driven by an external magnetic force in RBC suspensions. We applied the BEM to solve the fluid dynamics and a finite element method (FEM) to deal with the solid mechanics of the RBC membrane; the dynamic motions of the robot in blood were accurately calculated. To quantify the effect of RBCs on robot migration and aid in the design of highly efficient propulsion systems, we calculated the relative resistance coefficient of the robot ($C_r^*$). Furthermore, to extend the relevance of our findings and make them applicable across a wider range of microrobot geometries, we also explore the relative resistance coefficients of prolate ellipsoidal microrobots along its long and short axes, respectively. Although we have confirmed that our RBC model can reproduce experimental results such as dynamic modes in shear flow[50] and apparent viscosity[46], it has not been validated against robot locomotion. To bridge this gap, we additionally conducted experiments using pig blood to validate our numerical results. Additionally, machine learning (ML) is applied over wide ranges of the above parameters to predict the robot resistance coefficient of a spherical robot. The numerical results show that $C_r^*$ is significantly affected by hematocrit levels ($\phi$). The $C_r^*$ values obtained from simulations and experiments exhibit strong agreement across various $\phi$ levels. Additionally, $C_r^*$ is smaller than the macroscopic relative apparent viscosity of RBC suspensions under shear flow. Robot to RBC radius ratios ($\epsilon$) also obviously affect $C_r^*$ values, whereas the ratio of the driving magnetic force to the elastic force acting on the RBC membrane (dimensionless magnetoelastic number, $\Gamma$) does not have a noticeable impact. Moreover, the aspect ratio of a prolate ellipsoidal microrobot significantly affects the relative resistance coefficient along the long-axis direction. Furthermore, We present a high-accuracy model to predict $C_r^*$ using the extreme gradient boosting (XGBoost) algorithm. These findings provide a foundation for the design and optimization of microrobot propulsion in vivo.

## Results and Discussion

### Robot locomotion

We first calculated the displacement **d** of a single robot navigating in a quiescent RBC suspension (unless otherwise stated, all simulations were performed in quiescent RBC suspensions.) in $x$, $y$, and $z$ directions, as shown in Fig. 1. Unless otherwise specified, all results presented are from simulations. The propulsion of the robot is achieved through a magnetic thrust force ($F_m$) directed along the negative $z$-axis. It is assumed that the RBCs exhibit neutral buoyancy within the suspension. The displacement in the $z$ direction is much greater than in the $x$ and $y$ directions, attributed to the application of the magnetic thrust force along the negative $z$-axis. The horizontal velocity is generated by interactions with RBCs, which is sufficiently small to render horizontal robot spreading negligible. In all simulations, the horizontal velocity component is smaller than 4.1% of the velocity component in the $z$ direction. Hence, we used the velocity component in only the $z$ direction ($v_z$) to calculate the robot resistance coefficient. The drag coefficient is always considered in scenarios with high Reynolds numbers. In contrast, a resistance matrix is more appropriate when Stokes flow is present. To avoid confusion, a dimensionless quantity that we refer to as the resistance coefficient is used to quantify robot drag in RBC suspensions. As the drag is equal to the $F_m$ in Stokes flow, the resistance coefficient $C_r$ is as follows:

$$C_r = F_m/\mu a_s v_z. \tag{1}$$

The resistance coefficient relative to that without RBCs is calculated as:

$$C_r^* = \frac{C_r}{6\pi} = \frac{v_N}{v_z} \tag{2}$$

where $6\pi$ is the resistance coefficient in a Newtonian fluid (without RBCs) and $v_N$ is the speed in a Newtonian fluid (without RBCs), as given by Eq. (8).

### Effect of Hematocrit on the resistance coefficient

In this section, we change the hematocrit of quiescent RBC suspensions ($\phi$) to assess their impact on the robot resistance coefficient. Magnetoelastic number ($\Gamma$) and the size ratio between the robot and RBCs ($\epsilon$) are held constant at 1.0. To evaluate the difference between microscopic resistance coefficient and macroscopic relative apparent viscosity, we first calculated apparent viscosity $\eta$ of the pure RBC suspension, in the absence of the robot, at the macroscale as outlined in Supplementary Note 1 and Supplementary Fig. 1. Then, the relative apparent viscosity defined as $\eta^* = \eta/\mu$, which is critical for determining Stokes drag at the macroscale, is utilized to compared with $C_r^*$. Figure 2a shows the effect of hematocrit ($\phi$) on $C_r^*$ and $\eta^*$. Both $C_r^*$ and $\eta^*$ increase with increasing $\phi$. $\eta^*$ exhibits an increase as the $\phi$ grows in the semi-dilute regime, consistent with previous studies[53].

The maximum $\eta^*$ is 1.38 when $\phi$ is 25%. The maximum $C_r^*$ of a microrobot, on the other hand, is 1.23 at $\phi$ = 25%. $C_r^*$ is always lower than $\eta^*$. This is presumably because the microrobot is propelled through the fluid among RBCs, and the fluid has a viscosity lower than the macroscopic apparent viscosity of the RBC suspensions. Most current medical and surgical robots have millimeter-scale actuators, which are significantly larger than red blood cells. When they operate in the human body, the blood's rheological properties are typically treated in terms of apparent viscosity, considering it as a continuous and homogeneous fluid. The interaction between those large robot and RBCs can be likened to macro-rheological properties of blood. However, our study demonstrates the deficiencies of this macro-rheological approach when the robot size is comparable to that of red blood cells, such as with microrobots. The discrepancy is due to the different physical nature of the dimensionless resistance coefficient and the dimensionless apparent viscosity. The latter considers the interaction of

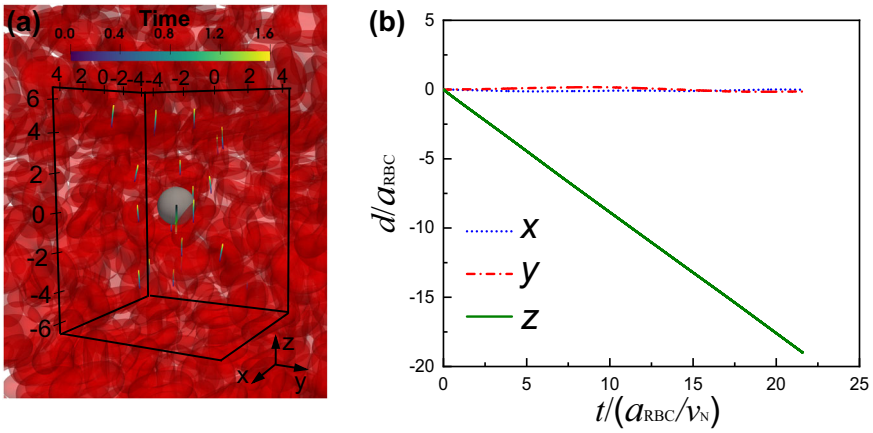


**Fig. 1 | A robot in an RBC suspension. a** Schematic of a robot propelled through an RBC suspension with $\phi$=20% and $\Gamma = \epsilon = 1.0$. The robot is the gray sphere. The colored lines show the trajectories of the robot and the centers of 19 RBCs in the main domain at $0 \le t/(a_{RBC}/v_N) \le 1.6$. **b** The displacement of the robot in the $x$ (blue dots), $y$ (red dash-dot line), and $z$ (green line) directions.

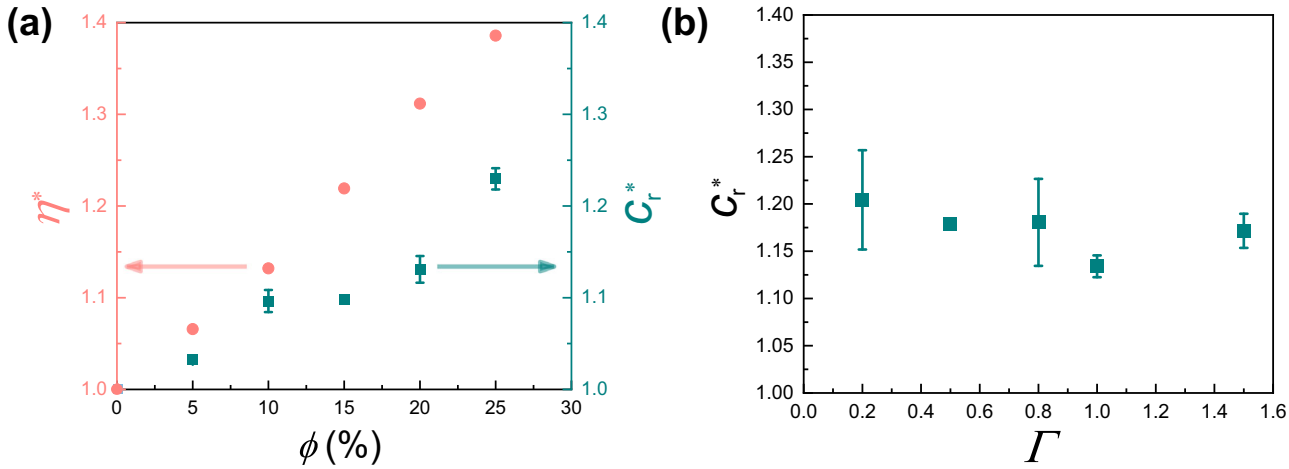


**Fig. 2 | Effect of hematocrit ($\phi$) and magnetoelastic number ($\Gamma$) on robot locomotion in RBC suspensions. a** Effect of $\phi$ on the relative resistance coefficient of the robot ($C_r^*$) in RBC suspensions (dark green) and relative apparent viscosity ($\eta^*$) of pure RBC suspensions (pink). $\Gamma$, $\epsilon$ and capillary number Ca were set to 1.0. Identical values of $\Gamma$ and Ca are employed to ensure that RBC deformation becomes similar around the robot and in the pure RBC suspension (see Supplementary Note 1). **b** Effect of $\Gamma$ on $C_r^*$ in an RBC suspension ($\phi$ = 20% and $\epsilon$ = 1.0). Each data point corresponds to robot locomotion from three independent trials. Error bars correspond to the standard deviation and are not visible when they are smaller than the data points.

RBCs alone, whereas the former considers the interaction between RBCs and a robot. These results clearly indicate the effect of robot size on drag in RBC suspensions.

The increasing tendency is also different from that in dilute suspensions of hard spheres. In such suspensions, the relative apparent viscosity is calculated using the Einstein equation:

$$\eta_d^* = 1 + 2.5\phi_r. \tag{3}$$

The speed of a sphere sedimenting in dilute suspensions of hard spheres ($v_{N,d}$) has been derived by Batchelor & Wen[41,42] and Reed & Anderson[43] as

$$v_{N,d} = v_N(1 + S_d\phi_r), \tag{4}$$

where $S_d$ is a speed correct coefficient, and $S_d$ is nearly a constant at -2.5 for a dilute suspension of neutrally buoyant spheres (Supplementary Table 1). Hence, the relative resistance coefficient of a sphere ($C_r^{*,s}$) at the dilute limit is:

$$C_r^{*,s} = \frac{v_N}{v_{N,d}} = \frac{1}{1 - 2.5\phi_r} > 1 + 2.5\phi_r = \eta_d^*. \tag{5}$$

The above shows that $C_r^{*,s}$ is always higher than $\eta_d^*$ in the dilute regime, in contrast to the scenario of Fig. 2a. There are two possible reasons for this difference: RBC deformability and the relatively high hematocrit level. When the robot navigates near RBCs, the cells may deform such that there is less resistance to the robot. In addition, a relatively high hematocrit level makes the suspension microstructure more complex. To further explore this, we simulated a robot driven by the same magnetic force in dilute suspensions as spherical capsules with a reduced $\Gamma$ in Supplementary Note 2 and Supplementary Fig. 2; the capsules did not deform and thus behaved like hard spheres. The $S_d$ approaches -2.5, which is very close to that for dilute suspensions of hard spheres[41–43]. Thus, our simulation method is reliable. Note that, in Fig. 2a, the gap between $C_r^*$ and $\eta^*$ is clearly smaller in more dilute RBC suspensions. These results indicate that the difference described above is explained by both RBC deformability and the relatively high hematocrit level. We conclude that the drag of the robot in RBC suspensions is smaller than that imparted by the macroscopic apparent viscosity of RBC suspensions.

### Effect of magnetoelastic number on the resistance coefficient

In this section, we investigated the effect of magnetoelastic number $\Gamma$ on the relative resistance coefficient of the robot ($C_r^*$) in quiescent RBC suspensions, by changing only $\Gamma$ ($\phi$ = 20% and $\epsilon$ = 1.0). The results are shown in Fig. 2b. Variations in $\Gamma$ triggered only slight fluctuations in $C_r^*$; all values were around 1.1–1.25, indicating that robot drag is not strongly affected by the deformability of RBCs in this $\Gamma$ regime.

### Effect of radius ratio on the resistance coefficient

As robot size greatly affects drag, the size ratio between the robot and RBCs ($\epsilon$), which is given by Eq. (10), was tuned to explore the effect thereof on the relative resistance coefficient of the robot ($C_r^*$) in quiescent RBC suspensions. The hematocrit level ($\phi$) and magnetoelastic number ($\Gamma$) were set to 20% and 1.0, respectively. The results are shown in Fig. 3a. Note that $C_r^*$ plateaus when $\epsilon$ is less than 0.7, followed by a sharp decline from followed by a sharp decline from $\epsilon$ = 0.7 to $\epsilon$ = 1.3, and eventually a slight increase with an enhanced $\epsilon$; this indicates that a smaller robot unfolds at higher $C_r^*$. This tendency is very different from sphere sedimentation in a dilute suspension of rigid and neutrally buoyant spheres reported by Batchelor[41,42]. The theoretical analysis of Batchelor revealed an independent effect of particle size on $C_r^*$. However, if neutrally buoyant (suspended) spheres are replaced by heavier spheres, the theoretical results show that smaller test spheres unfold at higher $C_r^*$ values in dilute suspensions. Based on the prediction using ML technology (Supplementary Note 3 and Supplementary Fig. 3), we found that the effect of $\epsilon$ was weak at $\phi \leq$ 10%, and could be ignored at $\phi \leq$5% when considering robot locomotion in RBC suspensions.

To investigate the mechanism underlying the disparity in $C_r^*$, we conducted additional simulations involving a single RBC situated directly beneath a robot. Periodic boundary conditions were not applied in this scenario. The schematic is depicted in the insets of Fig. 3b. The initial separation between the RBC and the microrobot was established at ($\epsilon$ + 1.8) $a_{RBC}$ in $z$ direction, and the angles ($\theta_{RBC}$) between the normal vector of the RBC plane ($\mathbf{n}_{RBC}$) and $z$-axis was varied. All other settings remaining unchanged. At $\theta_{RBC} = 0^\circ$, a slight separation was introduced between the slowest robot center and the RBC center in $x$ direction. We calculated the $C_r^*$ from the initial moment until the robot fully traversed the RBC, employing same methodology outlined in Eq. (2). For the big robot with $\epsilon$ =1.9, $C_r^*$ is nearly kept a constant of 1.0 with varying $\theta_{RBC}$. On the contrary, For the small robot with $\epsilon$ =0.4, $C_r^*$ significantly decreases as $\theta_{RBC}$ increases from 0° to 40°, then exhibits a gradual decrease after $\theta_{RBC} = 40^\circ$. The discrepancy aligns with observed robotic behaviors in RBC suspensions, indicating that smaller robots are prone to entrapment within red blood cells, whereas larger robots navigate through RBCs with relative ease.

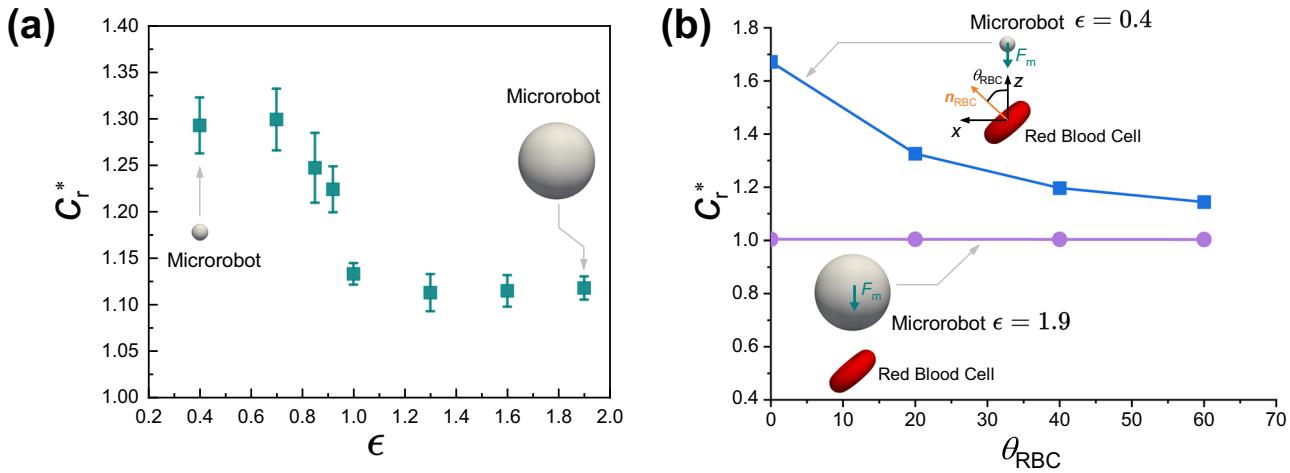


**Fig. 3 | Effect of radius ratio between the robot and RBCs ($\epsilon$) on robot locomotion. a** Effect of $\epsilon$ on the relative resistance coefficient of the robot ($C_r^*$) in an RBC suspension ($\phi$ = 20% and $\Gamma$ = 1.0). Each data point corresponds to robot locomotion from three independent trials. Error bars correspond to the standard deviation. **b** Variation of $C_r^*$ under the influence of a solitary RBC, with $\epsilon$ = 0.4 (blue) and $\epsilon$ = 1.9 (violet). The insets show the simulation setup, including the adjustment of the angle ($\theta_{RBC}$) between the normal vector of the RBC plane ($\mathbf{n}_{RBC}$) and $z$-axis. $C_r^*$ is calculated from the initial moment until the slowest robot fully traversed the RBC, employing same methodology outlined in Eq. (2).

## Resistance coefficient of ellipsoidal robot

Beyond spherical shapes, robots frequently exhibit intricate geometries, including ellipsoidal shapes, helical shapes, and unconventional forms. Consequently, the relative resistance coefficient varies in terms of the orientation and locomotion direction of the robot. Drag on a robot with a complex elongated shape (like a screw or filament) can be well approximated via resistive force theory (RFT)[54–56]. In RFT, the robot body is subdivided into smaller segments, and the linear superposition of forces acting on these segments throughout the robot body enables the prediction of robotic swimming velocities and efficiencies. Estimating the force acting on an segment involves utilizing the resistance coefficients of elongated particles and their velocities, with the former being predictable through the resistance coefficients of prolate ellipsoidal particles.

In this section, we investigated the relative resistance coefficients of the prolate ellipsoidal robots in two orientations based on its long axis: parallel ($C_{r//}^*$) to and perpendicular ($C_{r\perp}^*$) to the magnetic force $\mathbf{F}_m$ directed along negative $z$-axis. To maintain these orientations, a sufficiently high magnetic torque ($\mathbf{T}_m$) was applied to each robot as follows:

$$\mathbf{T}_m = V_e|\mathbf{M}||\mathbf{B}|\sin(\theta)\mathbf{n}_T = T_m \sin(\theta)\mathbf{n}_T \quad (6)$$

where $V_e$ and $\mathbf{M}$ represent the volume and the magnetization of the prolate ellipsoidal robot, respectively. In the analysis of $C_{r//}^*$, the magnetization vector $\mathbf{M}$ is aligned with the long axis of the robot. Conversely, for $C_{r\perp}^*$, $\mathbf{M}$ aligns with the short axis of the robot. $\mathbf{B}$ denotes the external magnetic field along negative $z$-axis, and $\theta$ is the angle between $\mathbf{M}$ and $\mathbf{B}$. $\mathbf{n}_T$ is the unit vector associated with the magnetic torque. The maximum magnitude of $\mathbf{T}_m$ is labeled $T_m$, set at $1000\mu U_0 a_s^2$ for the above-mentioned purpose. In the simulations discussed in this section, $\theta$ was observed to be nearly zero, indicating the maintenance of the specified orientations for the robots. The hematocrit level ($\phi$) and magnetoelastic number ($\Gamma$) were set to 20% and 1.0, respectively. The volume of each prolate ellipsoidal robot ($V_e$) was adjusted to match that of the spherical robot with $\epsilon$ = 1.0, while the aspect ratio of the prolate ellipsoidal robots ($\zeta$) was tuned. The velocities of the prolate ellipsoidal robot were simulated both in the absence and presence of RBCs for the calculation of $C_{r//}^*$ and $C_{r\perp}^*$, following the procedure described in Eq. (2). The results are shown in Fig. 4. $C_{r//}^*$ and $C_{r\perp}^*$ are nearly identical when $\zeta < 2.0$. $C_{r//}^*$ declines when $\zeta \geq 2.0$, while $C_{r\perp}^*$ exhibits only slight fluctuations with varying $\zeta$. This implies that prolate ellipsoidal robots with higher aspect ratios are more significantly affected by RBCs along the long-axis direction, but the aspect ratio has little impact on the relative resistance coefficient along the short-axis direction. The reason for the lower resistance of higher aspect ratios when moving along the long axis can be attributed to the effect of the more pointy tip colliding with the RBC, which deforms it and opens up the pathway.

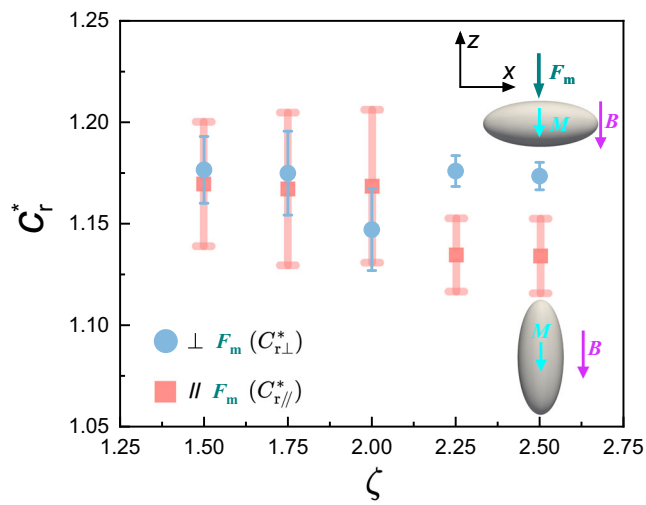


**Fig. 4 | Effect of aspect ratio of the prolate ellipsoidal robot ($\zeta$) on its relative resistance coefficient in an RBC suspension.** The volume of the prolate ellipsoidal robot was modified to align with that of the spherical robot with $\epsilon$ = 1.0. $\phi$ and $\Gamma$ are set at 20% and 1.0, respectively. Magnetic torque is applied to keep the orientation of the prolate ellipsoidal robot. Magnetic thrust force is always along the negative $z$-axis. Each data point corresponds to robot locomotion from three independent trials. Error bars correspond to the standard deviation.

## Experimental Results

To assess the reliability of our simulation, complementary experiments were conducted, followed by a comparison between experimental and simulation results. $\Gamma_{exp}$, calculated in terms of experimental parameters, was determined to be 0.14. These experimental parameters include $a_s$ = 2.5 μm, $\rho_p$ (particle density) = 7.87 g cm$^{-3}$, $\rho$ (density of ambient fluid)= 1.0 g cm$^{-3}$, $a_{RBC}$ = 3.0 μm reported in[57], and $G_s$ = 2.5 μN m$^{-1}$ reported in[52,58]. We employed the same methodology outlined in Eq. (2) to calculate the relative resistance coefficient ($C_r^*$). In the control group, $v_N$ exhibits a linear

relationship with $a_s^2$, following Stokes' drag equation. We obtained all values of $v_N$ for particles within the diameter range from 4 μm to 6 μm through a linear fitting based on ten independent experiments. Finally, we obtained $C_r^*$ by comparing $v_z$ of a particle from the test RBC suspension with the fitted $v_N$ value for a particle of the same size. The corresponding simulation was executed with $\Gamma = 0.14$, $\epsilon = 0.83$, and varying hematocrit levels $\phi$ (25%, 15%, and 5%). The experimental and simulated results are shown in Fig. 5(a). Welch's t-test reveals significant disparities in $C_r^*$ across varying $\phi$ for experimental results, marked by $P$ values of less than 0.001 (***), or less than 0.05 (*). These $P$ values highlight a $\phi$-dependent influence on $C_r^*$.

The simulated results are consistent with the experimental findings. $C_r^*$ increases as $\phi$ becomes more pronounced. Wider data distribution observed in experiments compared to simulations may be attributed to two factors: counting particles in an interval from 4 μm to 6 μm in experiments and potential inconsistencies in chamber heights caused by manual assembly. Essentially, the congruence between our simulation results and experimental findings serves as robust validation for the accuracy and reliability of our computational model and methodology. This validation bolsters our confidence in utilizing this simulation framework for further explorations of and insights into the behavior of a microrobot in RBC suspension.

## Machine learning

Simulations of large numbers of RBCs are very time-consuming and computationally expensive, rendering it difficult to obtain the relative resistance coefficient of the robot ($C_r^*$) when the hematocrit level ($\phi$), magnetoelastic number ($\Gamma$), and size ratio between the robot and RBCs ($\epsilon$) vary simultaneously. Machine learning (ML) utilizing various algorithms and models, such as extreme gradient boosting (XGBoost), random forest, support vector machine, and Naive Bayes models, greatly aids $C_r^*$ predictions[59–61]. XGBoost is an ensemble model based on decision trees that are individually trained; the results are then combined. XGBoost is appropriate for handling small datasets[59,60,62,63].

A flow chart of ML using XGBoost is shown in Fig. 6a. First, we simulated spherical robot locomotion at various $\phi$, $\Gamma$, and $\epsilon$ values, and calculated the corresponding $C_r^*$ values for 80 cases. Unlike the simulations in previous sections for which we derived mean values, each case was simulated only once given the time and computational costs. The data were then randomly divided into a training set (80%) to develop the model and a test set (20%) to evaluate model performance. For the training set, we employed $k$-fold cross-validation ($k$=5) to enhance the accuracy and generalization capacity of XGBoost[59,64]. Hyperparameter optimization from grid search was performed (Supplementary Table 2), followed by model

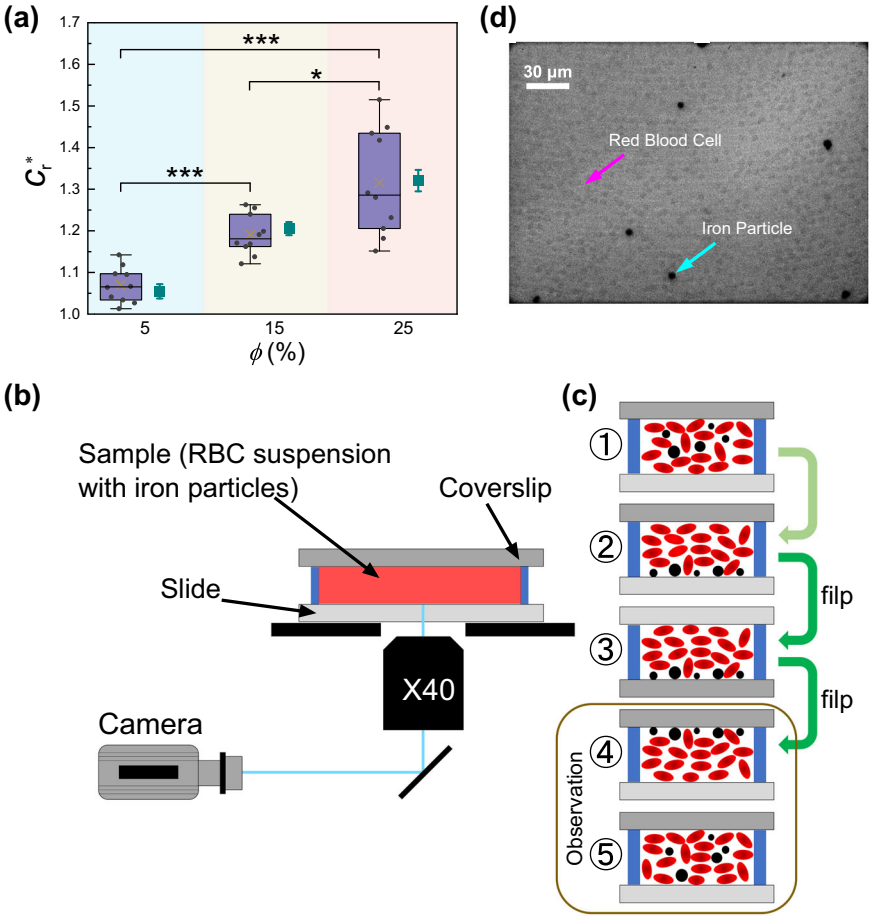


**Fig. 5 | Sedimentation of iron particles in red blood cell (RBC) suspension with various hematocrit levels. a** Comparison between our experiments and simulations for the relative resistance coefficient ($C_r^*$) of the gravity-driven robot under various hematocrit levels ($\epsilon = 0.83$ and $\Gamma = 0.14$). Box plots display the experimental results (n=10 for each hematocrit level). Cross plots and center lines within the box plots denote mean and median values, respectively. Significant differences are indicated by $^*P < 0.05$ and $^{***}P < 0.001$ according to the Welch's $t$-test. Simulated results are represented using square symbols. Each data point corresponds to robot locomotion from three independent trials. Error bars correspond to the standard deviation. **b** Schematic of the experimental setup displaying a sample of RBC suspension containing iron particles, positioned between a slide and a coverslip. The imaging pathway features a 40× microscope objective leading to a camera. For enhanced visual distinction, the identical slide and coverslip, although identical, are depicted in different colors. **c** Sample processing steps: Initial state of the sample ① The sample is left undisturbed for thirty minutes, allowing particles to settle, after which the focal plane is carefully determined ② By inverting the chamber, particles are directed to settle onto the coverslip ③ Finally, the chamber is inverted for immediate observation ④ ⑤. **d** Microscopic view of the sedimentation at the chamber's bottom. The hematocrit level of the RBC suspension is 25%. Scale bar: 30 μm.

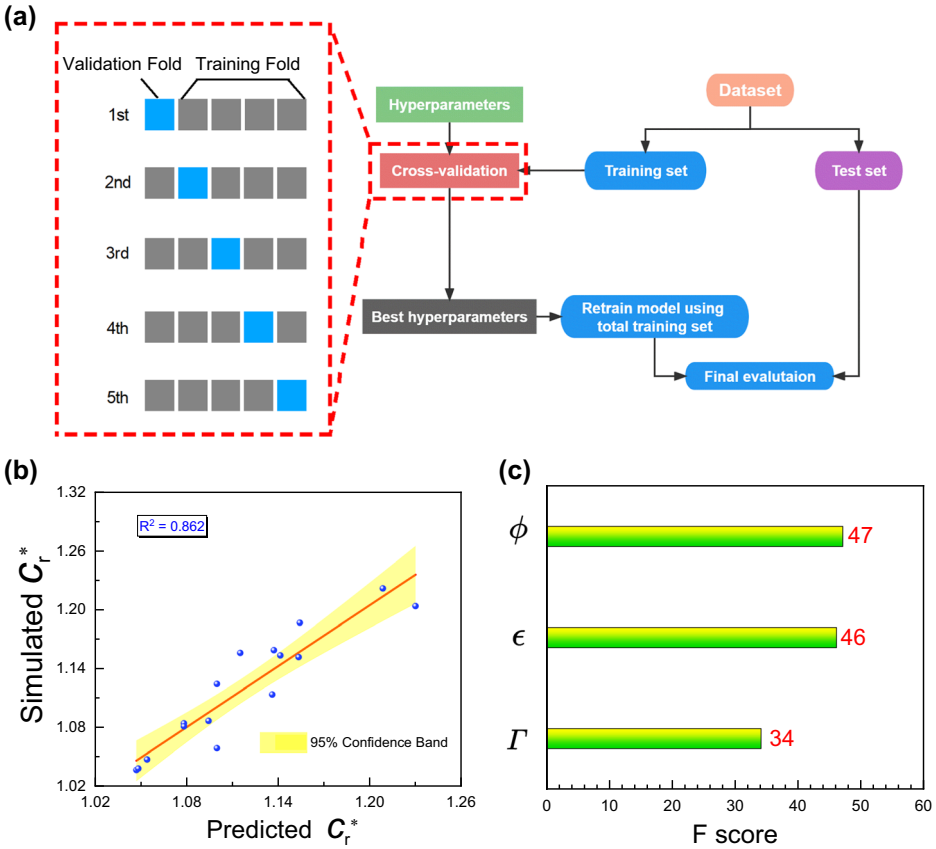


**Fig. 6 | Machine learning protocol for the relative resistance coefficient of the robot ($C_r^*$).** **a** Schematic flow chart of the ML protocol used to predict $C_r^*$. The zoomed view shows the cross-validation employed to determine the hyperparameters. **b** Simulated and predicted $C_r^*$. **c** Importance ranking of features affecting $C_r^*$ by summing the number of times each feature is used as a split node across all trees.

development using the optimized hyperparameters in the entire training set[65]. Finally, the coefficient of determination between the original and estimated data ($R^2$) of the test set was calculated for the XGBoost model; this is a measure of model performance. Fig. 6b shows linear fits between the values from simulations and ML predictions using XGBoost. The scatter points all lie close to the diagonal line, corresponding to a relatively high $R^2$ value of 0.862. Most points are included in the 95% confidence band, suggesting that the predicted and real values are reliably correlated. While employing a dataset of 64 data points may fall short of encapsulating the model's intrinsic characteristics, augmenting the data volume to 100 points yields only a slight improvement (Supplementary Note 4 and Supplementary Fig. 4a, b). Gaussian Process Regression (GPR), renowned for its adeptness at providing straightforward interpolation in high-dimensional data spaces and estimates of uncertainty in predictions, was employed for a comparative evaluation with the XGBoost method[66–68]. The $R^2$ value derived from the GPR-trained model is markedly low (Supplementary Note 4 and Supplementary Fig. 4c), suggesting that GPR struggles to encapsulate the complexity and non-linearity inherent in the function space, a challenge that XGBoost appears to navigate with greater efficacy.

The relationship between features ($\phi$, $\epsilon$, and $\Gamma$) and the relative resistance coefficient of the robot ($C_r^*$) were also derived by XGBoost in the form of an importance ranking (Fig. 6c). This ranking was calculated by summing the number of times each feature is used as a split node across all trees. Features used more frequently are considered to contribute more to $C_r^*$. Figure 6c shows that the hematocrit level ($\phi$) exhibits the highest score, i.e., exerted the greatest influence on $C_r^*$. The radius ratio between the robot and RBCs ($\epsilon$) also has a high value, suggesting that this also greatly affects $C_r^*$. The score for the magnetoelastic number ($\Gamma$) is low compared to the aforementioned features ($\phi$ and $\epsilon$), indicating a only modest effect on $C_r^*$. As discussed above, XGBoost makes highly accurate and interpretable predictions of the relative resistance coefficient of a robot driven by an external magnetic field in RBC suspensions. While XGBoost is initially introduced for predicting scenarios involving spherical robots, its adaptability extends seamlessly to robots with diverse geometries.

## Conclusion

In this paper, a robot driven by an external magnetic field in RBC suspensions was investigated via numerical simulations by coupling BEM and FEM. To assess the reliability of our simulation, we compared our experimental results with simulated results. The results match very well, indicating that the sufficient accuracy and reliability of our computational model and methodology. The relative resistance coefficient of the robot ($C_r^*$) was calculated at various hematocrit levels ($\phi$), robot to RBC radius ratios ($\epsilon$), and and driving magnetic force (acting on the robot) to elastic force (acting on RBCs) ratios ($\Gamma$). To generalize our model to a wider range of robot geometries, we explored prolate ellipsoidal robots with various aspect ratios in RBC suspensions. ML was used to predict $C_r^*$ within the parameter space of the above three features ($\phi$, $\epsilon$, and $\Gamma$). The main findings are as follows:

First, the hematocrit level ($\phi$) is the primary factor affecting the relative resistance coefficient of the spherical robot ($C_r^*$). $C_r^*$ becomes more pronounced with increasing $\phi$. A robot in an RBC suspension experiences less drag than the macro-drag calculated using the apparent viscosity. The drag is reduced presumably because the microrobot is propelled through the fluid among RBCs, and the fluid has a viscosity lower than the macroscopic apparent viscosity of the RBC suspensions. The situation differs from that in dilute suspensions of hard spheres; this is the first description thereof. Second, the robot to RBC radius ratio ($\epsilon$) affects $C_r^*$ to a lesser extent than the hematocrit level. In concentrated RBC suspensions, $C_r^*$ decreases sharply when $\epsilon$ increases from 0.7 to 1.3. However, in dilute RBC suspensions, this size effect can be ignored. Third, the ratio of the driving magnetic force acting on the robot to the elastic force acting

on RBCs ($\Gamma$) barely impacts $C_r^*$ in the parameter range employed here. Fourth, for the prolate ellipsoidal robots, $C_{r//}^*$ is influenced by the aspect ratio, whereas $C_{r\perp}^*$ exhibits minimal sensitivity to changes in the aspect ratio. Finally, the XGBoost can be used to establish a reliable dataset of $C_r^*$ values at the various $\phi$, $\epsilon$ and $\Gamma$ values.

Our simulation explored robot locomotion in blood and revealed the mechanism underlying the interaction between the robot and RBCs. Although we focused on a robot driven by an external magnetic field, robots driven by electric and optical fields, and chemicals, would be expected to behave similarly. Furthermore, our model can be adapted for other in vivo environments filled with cells, such as diseased cerebrospinal fluid containing various white blood cells. We plan to expand our research to account for wall effects, thereby simulating robot locomotion within blood vessels and capillaries.

## Methods

### Problem statement and basic equations

Consider a rigid spherical robot driven by an external magnetic force along the -$z$ direction in an infinite suspension of RBCs modeled as $N$ RBCs suspended in a triply periodic suspension. The magnetic thrust force $F_m$ imparted by a magnetic field $\mathbf{B}$ existing in $z$ direction only, written as:

$$F_m = \frac{4\pi}{3} a_s^3 |(\mathbf{M}\cdot\nabla)\mathbf{B}|, \tag{7}$$

where $a_s$ is the radius of the robot and $\mathbf{M}$ is the magnetization. We assume that the density of the robot is the same as that of the ambient fluid ($\rho$), and do not consider sedimentation effects. The ambient fluid is a Newtonian liquid of viscosity $\mu$. Given the small size of the robot, the surrounding flow is Stokes flow. As the effect of inertia is negligible, the Stokes' drag force and the magnetic thrust $F_m$ are balanced. The speed of the robot ($v_N$) in a Newtonian fluid of viscosity $\mu$ can be derived as:

$$v_N = \frac{2a_s^2|(\mathbf{M}\cdot\nabla)\mathbf{B}|}{9\mu}. \tag{8}$$

Note that this formula cannot be used when RBCs are present because it is assumed that the ambient fluid is a homogeneous fluid of viscosity $\mu$. Each RBC is modeled as a capsule with a hyper-elastic membrane. We assume that the internal and external RBC fluids have the same density $\rho$ and viscosity $\mu$. RBC deformation becomes more evident as the magnetic thrust of the robot increases. The ratio between the magnetic force and elastic force of the RBC membrane, $\Gamma$, is an important dimensionless magnetoelastic number that governs RBC deformation and is defined as:

$$\Gamma = \frac{a_s^3|(\mathbf{M}\cdot\nabla)\mathbf{B}|}{G_s a_{RBC}}, \tag{9}$$

where $G_s$ is the shear elastic modulus of RBC membrane and $a_{RBC}$ is the characteristic radius of an RBC defined as $\sqrt[3]{\frac{3V_R}{4\pi}}$, where $V_R$ is the volume of an RBC. Another important dimensionless number is $\epsilon$, which is defined as:

$$\epsilon = \frac{a_s}{a_{RBC}}, \tag{10}$$

which determines the size effect of the robot relative to that of RBCs.

When a robot navigates in an infinite RBC suspension (RBCs are assumed to be neutrally buoyant), i.e., a suspension with $N$ RBCs in a triply periodic flow field, the the velocity at a given point $\mathbf{x}$ is given by the boundary integral formulation[69]:

$$\begin{aligned}\mathbf{v}(\mathbf{x}) = &-\frac{1}{8\pi\mu}\sum_{m=1}^{N}\int \boldsymbol{J}^E(\mathbf{x},\mathbf{y})\cdot\mathbf{q}_R(\mathbf{y})\mathrm{d}A_{R,m}(\mathbf{y})\\ &-\frac{1}{8\pi\mu}\int \boldsymbol{J}(\mathbf{x},\mathbf{y})\cdot\mathbf{q}_s(\mathbf{y})\mathrm{d}A_s(\mathbf{y})\\ &+\frac{1}{8\pi\mu}\sum_{j=1}^{N_F}\boldsymbol{J}^E(\mathbf{x},\mathbf{y}_j^G)\cdot\mathbf{F}_{rep}(\mathbf{y}_j^G),\end{aligned} \tag{11}$$

where $A_R$ is the RBC surface and $A_s$ is the robot surface. $\mathbf{q}_R$ and $\mathbf{q}_s$ are the force densities on the RBC membrane and surface of the robot, respectively. $\boldsymbol{J}$ is the Green's function, given by:

$$J_{ij} = \frac{\delta_{ij}}{r} + \frac{r_i r_j}{r^3}, \tag{12}$$

where $\boldsymbol{\delta}_{ij}$ is the Kronecker delta, $\mathbf{r} = \mathbf{x} - \mathbf{y}$ is the vector from the source point $\mathbf{y}$ to the observation point $\mathbf{x}$, and $r = |\mathbf{r}|$. $\boldsymbol{J}^E$ is Green's function for the triply periodic lattice, which is solved using the Ewald summation technique, written as[70]:

$$J_{ij}^E = \sum_{\gamma} J_{ij}^I(\mathbf{r}_\gamma) + \frac{8\pi}{V}\sum_{\kappa\neq 0} J_{ij}^{II}, \tag{13}$$

$$J_{ij}^I = \frac{\delta_{ij}}{r}B_1 + \frac{r_i r_j}{r^3}B_2, \tag{14}$$

$$J_{ij}^{II} = B_3 \exp\left(-\frac{k^2}{4\xi^2}\right)\cos(\mathbf{k}\cdot\mathbf{r}), \tag{15}$$

$$B_1 = \mathrm{erfc}(\xi r) + \frac{\exp(\xi^2 r^2)}{\sqrt{\pi}}(4\xi^3 r^3 - 6\xi r), \tag{16}$$

$$B_2 = \mathrm{erfc}(\xi r) + \frac{\exp(\xi^2 r^2)}{\sqrt{\pi}}(2\xi r - 4\xi^3 r^3), \tag{17}$$

$$B_3 = \left(\frac{\delta_{ij}}{k^2} - \frac{k_i k_j}{k^4}\right)\left(1 + \frac{k^2}{4\xi^2} + \frac{k^4}{8\xi^4}\right), \tag{18}$$

$$\mathbf{k} = \left(\frac{2\pi n_x}{l_x}, \frac{2\pi n_y}{l_y}, \frac{2\pi n_z}{l_z}\right), \tag{19}$$

where $l_x$, $l_y$, and $l_z$ are the lattice sizes in the $x$, $y$ and $z$ directions, respectively, and $\xi = \pi^{\frac{1}{2}}(l_x l_y l_z)^{-\frac{1}{3}}$. $\gamma$ is the index of periodic boxes and $\kappa$ is the index of reciprocal vectors. $\mathbf{k}$ is the reciprocal vector and $k = |\mathbf{k}|$. $n_x$, $n_y$, $n_z$ are the integer numbers of the indices of the periodic boxes. To avoid RBC-RBC and RBC-robot overlaps, a very short-range repulsive force $\mathbf{F}_{rep}$ is employed, which acts on the observation Gauss point $\mathbf{y}_j^G$ and is expressed as:

$$\mathbf{F}_{rep}(\mathbf{y}_j^G) = \begin{cases} \sum_i^m k_c d_G \frac{\mathbf{y}_j^G - \mathbf{x}_i^G}{|\mathbf{y}_j^G - \mathbf{x}_i^G|} & d_G > 0 \\ 0 & d_G \le 0 \end{cases} \tag{20}$$

where $k_c$ is a spring constant and $d_G$ is defined as:

$$d_G = 0.18a_{RBC} - \left|\mathbf{y}_j^G - \mathbf{x}_i^G\right|, \tag{21}$$

$\mathbf{x}_i^G$ are the nearby Gauss points of distinct RBCs. In this study, $k_c$ is set to $0.04G_s$ and the repulsive force is considered only when the distance between two Gauss points on different RBCs is smaller than $0.18a_{RBC}$.

The RBC membrane is assumed to be an isotropic and hyperelastic material that follows the Skalak law[45]. The surface deformation gradient tensor $\boldsymbol{D}_s$ is given by:

$$d\mathbf{x} = \boldsymbol{D}_s \cdot d\mathbf{X}, \tag{22}$$

where $\mathbf{X}$ and $\mathbf{x}$ are the membrane material points of the reference and deformed states, respectively. Local deformation of the RBC membrane is described by the Green-Lagrange strain tensor:

$$\boldsymbol{E} = \frac{1}{2}(\boldsymbol{D}_s^T \cdot \boldsymbol{D}_s - \boldsymbol{I}_s), \tag{23}$$

where $\boldsymbol{I}_s$ is the tangential projection operator. Two invariants of the in-plane strain tensor $\boldsymbol{E}$ are:

$$I_1 = \lambda_1^2 + \lambda_2^2 - 2, \tag{24}$$

$$I_2 = \lambda_1^2\lambda_2^2 - 1, \tag{25}$$

where $\lambda_1$ and $\lambda_2$ are the principal stretch ratios. The Cauchy stress tensor $\boldsymbol{T}$ is:

$$\boldsymbol{T} = \frac{1}{S}\boldsymbol{D}_s \cdot \frac{w_s}{\boldsymbol{E}} \cdot \boldsymbol{D}_s^T, \tag{26}$$

where $S = \lambda_1\lambda_2$ is the area dilation ratio. Invoking the two-dimensional constitutive law of Skalak, the elastic strain energy per unit area $w_s$ can be expressed as:

$$w_s = \frac{G_s}{2}(I_1^2 + 2I_1 - 2I_2 + CI_2^2), \tag{27}$$

where $C$ is the area dilation modulus (10 in this paper)[52].

### Numerical method

The numerical method developed by Walter et al.[45] was applied to couple the BEM of fluid mechanics to the FEM of solid mechanics. We used this method in previous papers[48–51]. A single RBC membrane is discretized using an unstructured triangular mesh produced via recursive subdivision of an icosahedron. In solid mechanics, the weak form of the equilibrium condition of the RBC is:

$$\int_A \hat{\mathbf{u}} \cdot \mathbf{q} dA = \int_A \hat{\boldsymbol{\epsilon}} : \mathbf{T} dA \tag{28}$$

which is solved by the FEM, where $\hat{\mathbf{u}}$ and $\hat{\boldsymbol{\epsilon}}$ are the virtual displacement and the virtual strain, respectively. In fluid mechanics, the BEM combined with multipole expansion is applied to solve Equation (11). The surface integrals of the BEM are computed using the Gaussian quadrature approach. Polar coordinates are utilized to eliminate the $1/r$ singularity[53,71]. The velocity of the RBC membrane at point $\mathbf{x}$ is given by the kinematic condition:

$$\frac{d\mathbf{x}}{dt} = \mathbf{v}(\mathbf{x}). \tag{29}$$

The position is updated using the explicit second-order Runge-Kutta method. The implementation of a volume constraint prevents even a modest volume error[71].

When resolving the velocity disruptions caused by near-field forces, a fine mesh (small element size) is vital; however, the velocity caused by far-field forces can be coarse-grained when using the adaptive mesh method. In the simulations, when a certain RBC (source set) is distant from another RBC or the robot (including the observation point), a patch of 16 elements is used to replace the triangular elements. Thus, an RBC is constructed from 80 large patches. In contrast, an RBC has 1,280 triangular elements when a fine mesh is required. The force density on a patch with $N_e$ fine elements ($q_c$) is given by:

$$q_c = \frac{1}{N_e}\sum_i^{N_e} q_f^i. \tag{30}$$

where $q_f^i$ is the force density exerted on fine mesh.

To enhance the stability of simulations, the second-layer computation of the Ewald summation was introduced. Unfortunately, this greatly increased the computational burden. To reduce this load while retaining accuracy, the boundary integral of $\mathbf{q}_R$ in Eq. (11) is expanded in terms of moments[72,73]:

$$\int \boldsymbol{J}^E(\mathbf{x},\mathbf{y}) \cdot \mathbf{q}_R(\mathbf{y}) dA_{R,m}(\mathbf{y}) = K_{ijk}^{E_I} H_{jk}^m + \frac{1}{V} K_{ijk}^{E_{II}} H_{jk}^m + H.O.T.\ , \tag{31}$$

where $V = l_x l_y l_z$, and $H.O.T.$ are the higher order terms. The force and torque terms disappear due to the force- and torque-free conditions. The stresslet $\boldsymbol{H}^m$ is as follows:

$$H_{ij}^m = \frac{1}{2}\int \left[\hat{r}_i q_j(\mathbf{y}) + \hat{r}_j q_i(\mathbf{y}) - \frac{2}{3}\delta_{ij}\hat{r}_k q_k(\mathbf{y})\right] dA_{R,m}(\mathbf{y}) \tag{32}$$

where $\hat{\mathbf{r}}$ is the vector from the center of a certain RBC to a defined point on the membrane of the same RBC. The propagators are:

$$\begin{aligned} K_{ijk}^{E_I} = & -\tfrac{1}{2}\left(\delta_{ij}\tfrac{r_k}{r} + \delta_{ik}\tfrac{r_j}{r}\right)\tfrac{\exp(-\xi^2 r^2)}{\sqrt{\pi}}(-8\xi^5 r^3 + 16\xi^3 r) \\ & -\delta_{jk}\tfrac{r_i}{r}\left[\tfrac{1}{r^2}\mathrm{erfc}(\xi r) + \tfrac{\exp(-\xi^2 r^2)}{\sqrt{\pi}}(-4\xi^3 r + \tfrac{2\xi}{r})\right] \\ & -\tfrac{r_i r_j r_k}{r^3}\left[-\tfrac{3}{r^2}\mathrm{erfc}(\xi r) + \tfrac{\exp(-\xi^2 r^2)}{\sqrt{\pi}}(8\xi^5 r^3 - 4\xi^3 r - \tfrac{6\xi}{r})\right], \end{aligned} \tag{33}$$

$$\begin{aligned} K_{ijk}^{E_{II}} = & \frac{4\pi}{k}\left[\frac{k_k}{k}(\delta_{ij} - \frac{k_i k_j}{k^2}) + \frac{k_j}{k}(\delta_{ij} - \frac{k_i k_k}{k^2})\right]\left(1 + \frac{k^2}{4\xi^2} + \frac{k^4}{8\xi^4}\right) \\ & \exp\left(-\frac{k^2}{4\xi^2}\right)\sin(\mathbf{k}\cdot\mathbf{r}). \end{aligned} \tag{34}$$

We used the stresslet as a substitute for an RBC during second-layer Ewald summation; this significantly accelerated the computation. The nomenclature and parameter ranges used in this study are summarized in Supplementary Table 3.

### Experimental method and materials

To validate the simulation, we conducted experiments involving the sedimentation of spherical particles in an RBC suspension. Similar to the magnetic force used in the simulation, gravity force is also considered a body force, and it can serve as a driving force for the particles. In this context, we denote $\Gamma$ as $\Gamma_{exp}$ and define it as follows:

$$\Gamma_{exp} = \frac{a_s^3(\rho_p - \rho)g}{G_s a_{RBC}}. \tag{35}$$

Here, $\rho_p$ and $g$ are the particle density and the gravitational acceleration. $\Gamma_{exp}$ still represents the ratio of driving force and elastic force acting on the RBC membrane, although the driving force has shifted from a magnetic force to a gravity force.

The preparation of the RBC suspension, following the same procedure as described in our previous research[74–76], is briefly outlined here. Pig blood was bought from Tokyo Shibaura Organ Corporation. The RBCs were separated from the bulk pig blood through centrifugation at 1500 rpm for 20 minutes (model 3220 Micro Hematocrit Centrifuge, Kubota). Subsequently, plasma and buffy coat were removed by aspiration. The washing and centrifuging with physiological saline (PS) were repeated twice. This process of washing and centrifugation with physiological saline (PS) was

repeated twice. The washed RBCs were then diluted with dextran 40 (DX-40, Otsuka Medicine)

The observation chamber was assembled using two cover glasses (NEO Micro cover glass; Thickness No. 1; 24 × 60 mm, Matsunami, Tokyo, Japan) and five layers of double-sided tape. The height of the chamber was approximately 1500 μm. Initially, the top cover glass, the observation chamber without the top cover glass, and iron particles (d50 = 5.0 μm, type REZL, Jiangsu Tianyi Ultrafine Metal Powder Co. Ltd., China) were immersed in bovine serum albumin (BSA, Stock Solution 130091-376, Miltenyi Biotech., Germany) for an hour to prevent particle aggregation and adhesion to pig RBCs or the inner surface of the observation chamber[77,78]. Subsequently, a mixture of the particle suspension and the RBC suspension was prepared to create the test suspension containing 25% RBCs (25% hematocrit), 15% RBCs, and 5% RBCs. The test suspension was injected into the observation chamber after BSA was aspirated. Finally, the observation chamber was sealed via the top cover glass. For the control group, the particle suspension and pure dextran 40 were mixed in the same proportion as in the test suspension. Consequently, all components in the test suspension and control suspension, except for RBCs, are identical.

Sample examination was performed using an inverted microscope (Olympus IX71; Olympus, Tokyo, Japan) equipped with a 40× magnification objective, as outlined in Fig. 5b. Initially, the test suspension or control suspension was placed on the microscope stage for 30 min, allowing the complete particle settlement (Fig. 5c ①②). The microscope's focus was then finely adjusted to yield a sharp image of the particles. Subsequently, the observation chamber was inverted and left in that position for an additional 30 minutes to direct particle settlement onto the coverslip (Fig. 5c ③). Finally, the observation chamber was inverted again after video recordings were captured using a high-speed camera operating at a frame rate of 50 fps (Fig. 5c ④⑤). For our analysis, only particles with diameters ranging from 4 μm to 6 μm were considered. A snapshot displaying the sedimentation at the chamber's bottom is shown in Fig. 5d. Sedimentation time could be calculated from the first frame that exhibits the stationary observation chamber to the frame where a distinct spherical iron particle becomes visible. The sedimentation velocity of particles can be calculated by dividing the depth of the chamber by the sedimentation time of the particles. Since the driving force, gravity, is constant, this sedimentation velocity can be compared to that without RBCs to obtain the relative resistance coefficient of a microrobot ($C_r^*$)

### Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

## Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

## Code availability

The code that supports the findings of this study is available from the corresponding authors upon reasonable request.

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## Author contributions
C. W. and T. I. designed research; T. O. and C. W. contributed analytic tools; C. W. performed research; C. W. and T. I. analyzed data; C. W., T. O. and T. I. wrote the paper.

## Competing interests
The authors declare no competing interests.

## Additional information
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